

Multimedia Application

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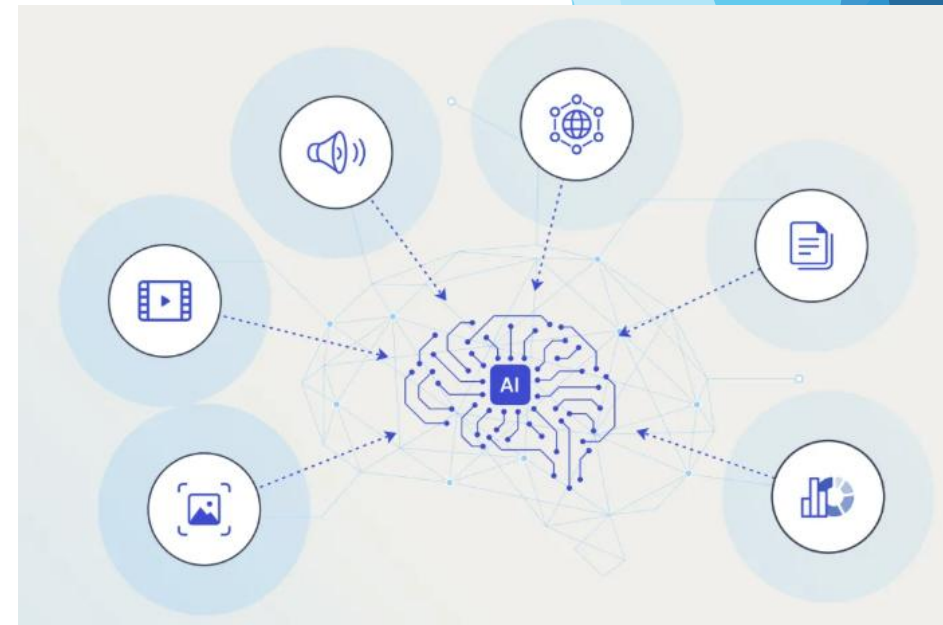
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Content

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- ▶ Key Characteristics of Multimodal AI
- ▶ Multimodal data types
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Multimodal Artificial Intelligence

- ▶ Multimodal Artificial Intelligence (AI) represents a paradigm shift in how machines process and understand information. Unlike traditional AI models that focus on a single data type, such as text or images, **multimodal AI systems are engineered to interpret and integrate diverse forms of data input, including text, images, audio, and video.**



Multimodal Artificial Intelligence

- ▶ This integrated processing allows AI to achieve a more nuanced and comprehensive understanding of the world, mirroring human cognition which naturally synthesizes information from multiple senses. As the pursuit of artificial general intelligence intensifies, the incorporation of multimodal capabilities is becoming increasingly vital for developing AI systems that can perform a broad spectrum of tasks with human-like proficiency.

Multimodal data types

- ▶ The core of **multimodal AI** lies in its ability to handle various **modalities of data**. A modality refers to a distinct type of data or sensory input. In the context of AI, common modalities include text, which encompasses written language; images, representing visual data; audio, capturing sound and speech; and video, integrating both visual and auditory information over time. The integration of these different modalities is crucial for tasks that require a holistic understanding.

Key Characteristics of Multimodal AI

- ▶ **Combines Multiple Data Types**
- ▶ **Enables More Sophisticated Interactions**
- ▶ **Replicates Human Brain's Multimodal Capabilities**
- ▶ **Intuitive and Natural Interactions**

Key Characteristics of Multimodal AI

- ▶ Multimodal AI systems can process and integrate information from various modalities, such as text, images, audio, and video.
- ▶ By combining different data types, multimodal AI can perform more complex tasks and make more accurate predictions.
- ▶ The human brain naturally integrates information from multiple senses, and multimodal AI aims to replicate this ability in machines.
- ▶ Multimodal AI can create more natural and intuitive user experiences by allowing users to interact with systems in a variety of ways, such as through voice commands, images, and text.

Example of multimodal system

- ▶ For instance, a multimodal system analyzing a video might process both the **visual content and the accompanying audio to provide a more accurate interpretation of the scene.**
- ▶ Unimodal AI models, in contrast, are restricted to processing a single type of input, which limits their applicability in scenarios where **information is inherently multimodal.** The ability to process and understand different sensory modes equips AI with enhanced reasoning, problem-solving, and generation capabilities, unlocking a wide array of possibilities for next-generation applications across various domains.

The landscape of multimodal AI

- ▶ The landscape of multimodal AI is populated by a growing number of sophisticated models, each designed to handle and integrate different combinations of modalities. These models leverage advancements in deep learning, particularly transformer architectures, to achieve state-of-the-art performance on various multimodal tasks. Some prominent examples include Meta AI's Llama 3.2 90B, which supports text and image inputs, Furthermore, models like ImageBind demonstrate the capability to integrate information from even more diverse sources, enhancing the model's overall understanding. These models are continually evolving, pushing the boundaries of what AI can perceive and comprehend.

Various models

Model	Modality Support
Llama 3.2 90B	Text, Image
Gemini 1.5 Flash	Text, Image, Video, Audio
Florence	Text, Image
GPT-4o	Text, Image, Audio (Input), Text (Output)
Claude 3.5	Text, Image
LLaVA V1	Text, Image
DALL·E 3	Text, Image
ImageBind	Multiple modalities
GPT-4	Text, Images, Audio (Input), Text (Output)

Examples of Multimodal AI in Action

- ▶ Smart Assistants
- ▶ Image Recognition and Captioning
- ▶ Natural Language Processing (NLP) with Context
- ▶ Customer Service

Examples of Multimodal AI in Action

- ▶ AI assistants like Google Assistant and Amazon Alexa can process voice commands, understand visual information from images and videos, and respond with both audio and text outputs.
- ▶ Multimodal AI models can analyze images, understand the content, and generate descriptive captions or summaries.
- ▶ Multimodal AI can combine text-based NLP with other modalities, like images or audio, to provide a more complete understanding of the context and meaning of a conversation.
- ▶ Multimodal AI can analyze customer feedback through text, voice tone, and facial expressions in videos to gain deeper insights into customer satisfaction and provide better support.

Applications of Multimodal AI

Content Creation:

- ▶ Multimodal AI can be used to generate creative content like images, videos, and text, by combining different modalities

Education:

- ▶ Multimodal AI can create more engaging and interactive learning experiences by combining different types of media

Healthcare:

- ▶ Multimodal AI can be used for **medical diagnosis**, by analyzing images, audio, and patient records

Security:

- ▶ Multimodal AI can be used for **advanced security systems** that analyze images, audio, and video to detect threats

Challenges and Future Directions

- ▶ Developing and deploying multimodal AI models presents several challenges :
- ▶ **Data Integration:** Effectively combining data from different modalities with varying characteristics.
- ▶ **Data Alignment:** Ensuring that data from different modalities corresponds to the same event or object.
- ▶ **Computational Resources:** Multimodal models often require significant computational power for training and inference.
- ▶ **Evaluation Metrics:** Defining appropriate metrics to evaluate the performance of models that process diverse data types.

Key Python Libraries for Multimodal AI

- ▶ **Hugging Face Transformers:** Provides access to thousands of pre-trained models, including many multimodal models, and simplifies their use for tasks like image captioning and visual question answering.
- ▶ **Weaviate:** An open-source vector database that allows for storing and searching multimodal embeddings, enabling cross-modal retrieval.
- ▶ **TensorFlow and Keras:** Powerful frameworks for building and training custom multimodal models, offering flexibility for complex architectures.
- ▶ **PyTorch:** Another popular deep learning framework with libraries like TorchMultimodal specifically designed for building and scaling multimodal models.

Reference

- ▶ <https://huggingface.co/models>
- ▶ Various Online sources

Question



Thank you

